

Investment Financing Conservatism and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Investment Financing Conservatism (IFC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on IFC achieves an annualized gross (net) Sharpe ratio of 0.59 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 3.18 (2.57), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth) is 20 bps/month with a t-statistic of 2.90.

1 Introduction

Asset pricing theory fundamentally concerns the relationship between risk and expected returns, with consumption-based models providing the theoretical foundation for understanding cross-sectional variation in equity returns. Despite decades of research establishing that systematic risk exposure should drive return differentials, significant puzzles remain regarding which firm characteristics capture economically meaningful variation in consumption risk. The challenge lies in identifying observable signals that reliably proxy for firms' exposure to aggregate consumption shocks and investors' marginal utility of wealth across different economic states. Investment Financing Conservatism (IFC), defined as the tendency of firms to rely on internal funds rather than external financing for investment activities, emerges as a potentially important characteristic that may capture firms' differential exposure to consumption risk. This paper examines whether IFC predicts cross-sectional stock returns through a consumption-based asset pricing lens, testing whether conservative financing policies reflect fundamental differences in how firms' cash flows covary with aggregate consumption dynamics.

The theoretical link between Investment Financing Conservatism and expected returns operates through firms' differential exposure to consumption risk across economic states. Firms exhibiting high IFC maintain financial flexibility by avoiding external capital markets, a behavior that becomes particularly valuable during periods of high marginal utility when consumption growth is low or negative (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). This conservative financing approach suggests these firms face lower systematic risk because their investment opportunities and cash flows are less sensitive to aggregate consumption shocks that drive the pricing kernel. The consumption-based asset pricing framework predicts that assets whose payoffs covary positively with consumption command lower risk premia, as they fail to provide insurance when marginal utility is highest (Cochrane, 2005;

Hansen and Singleton, 1982). During economic downturns characterized by low consumption growth, firms with high IFC can maintain operations and investment using internal resources, while firms dependent on external financing face severe constraints precisely when the stochastic discount factor is highest. This state-dependent variation in firms' exposure to consumption risk should manifest in the cross-section of expected returns. The long-run risks model further suggests that IFC may capture firms' differential exposure to persistent shocks to consumption growth, with conservative firms better positioned to smooth through temporary consumption fluctuations (Bansal and Yaron, 2004; Hansen et al., 2008). Moreover, in a habit formation framework, high IFC firms provide relatively stable cash flows that are less sensitive to the surplus consumption ratio, implying lower systematic risk and consequently lower expected returns (Campbell and Cochrane, 1999). The disaster risk perspective reinforces this mechanism, as financially conservative firms maintain options to deploy internal capital during rare disasters when marginal utility spikes and external financing becomes prohibitively expensive or unavailable (Barro, 2006; Gabaix, 2012).

Our empirical analysis reveals that Investment Financing Conservatism strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on IFC achieving an annualized gross Sharpe ratio of 0.59 and a monthly average return of 29 basis points with a t-statistic of 4.26. The strategy's performance remains robust after controlling for standard risk factors, generating a monthly alpha of 22 basis points relative to the Fama-French six-factor model with a t-statistic of 3.18. The economic magnitude of the IFC premium persists across different portfolio construction methodologies, with alphas ranging from 22 to 34 basis points per month depending on the factor model specification. Notably, the strategy loads significantly on the CMA (investment) factor with a coefficient of 0.29 (t-statistic of 6.28), consistent with IFC capturing variation in firms' investment behavior that re-

lates to consumption risk exposure. The IFC effect concentrates among larger stocks where implementation is most feasible, with the strategy earning 39 basis points per month among firms above the 80th NYSE size percentile (t-statistic of 4.43). After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains economically significant at 0.47, with the strategy generating net alphas ranging from 17 to 30 basis points across factor models. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related anomalies equals 20 basis points with a t-statistic of 2.90, demonstrating that IFC contains distinct information about consumption risk exposure not captured by existing predictors.

This paper contributes to the consumption-based asset pricing literature by identifying Investment Financing Conservatism as a powerful predictor of cross-sectional returns that captures firms’ differential exposure to consumption risk. Our work extends (Cochrane, 1991) and (Zhang, 2005) by demonstrating that financing conservatism, rather than investment levels alone, drives meaningful variation in expected returns through its connection to state-dependent consumption risk. Unlike (Cooper et al., 2008) who focus on asset growth as a mispricing signal, we show that IFC’s predictive power stems from rational risk compensation related to firms’ ability to maintain operations during high marginal utility states. Our findings complement (Hou et al., 2015) by revealing that the investment-return relation operates most strongly through firms’ financing choices rather than pure investment quantities, with IFC capturing unique variation beyond standard investment-based factors. Methodologically, we advance the literature by employing the comprehensive testing framework of (Novy-Marx and Velikov, 2023b), demonstrating that IFC’s performance survives stringent robustness tests including controls for 212 documented anomalies and transaction cost adjustments. The IFC strategy ranks in the top 3% of anomalies in terms of net Sharpe ratio, outperforming 98% of previously documented predictors after

implementation costs. Our results have important implications for understanding the consumption-based origins of the cross-section of expected returns, suggesting that firms’ financing conservatism serves as an observable proxy for their fundamental exposure to aggregate consumption risk. The finding that IFC maintains predictive power among large-cap stocks where arbitrage is most feasible strengthens the risk-based interpretation, as these stocks face fewer limits to arbitrage that could sustain mispricing. More broadly, our evidence supports consumption-based models that emphasize state-dependent risk premia and the importance of financial constraints in determining firms’ systematic risk exposures during periods of high marginal utility.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 4900-4999) and financial services companies (SIC codes 6000-6999) due to their unique capital structures and regulatory requirements. Investment Financing Conservatism signal requires two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s long-term debt obligations. Capital expenditure (CAPX) measures the funds used by a company to acquire, upgrade, and maintain physical assets such as property, plants, equipment, and technology infrastructure during the fiscal year. Both variables reflect important aspects of corporate financial policy—DLTIS captures financing decisions while CAPX represents investment ac-

tivities.construct the Investment Financing Conservatism signal by computing the year-over-year change in long-term debt issuance scaled by lagged capital expenditure: $(DLTIS(t) - DLTIS(t-1)) / CAPX(t-1)$. This ratio captures the relationship between changes in debt financing relative to the firm’s recent investment intensity. A higher value indicates that a firm is increasing its reliance on debt financing relative to its capital investment base, while a lower or negative value suggests more conservative financing behavior where debt issuance growth is modest or declining relative to capital spending. We calculate this measure annually using fiscal year-end values and require non-missing values for both DLTIS and CAPX in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where lagged CAPX is zero or negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFC signal. Panel A plots the time-series of the mean, median, and interquartile range for IFC. On average, the cross-sectional mean (median) IFC is -3.65 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input IFC data. The signal’s interquartile range spans -3.29 to 4.32. Panel B of Figure 1 plots the time-series of the coverage of the IFC signal for the CRSP universe. On average, the IFC signal is available for 6.07% of CRSP names, which on average make up 7.19% of total market capitalization.

4 Does IFC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high IFC portfolio and sells the low IFC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IFC strategy earns an average return of 0.29% per month with a t-statistic of 4.26. The annualized Sharpe ratio of the strategy is 0.59. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 6.28 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 504 stocks and an average market capitalization of at least \$1,502 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.18. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-25bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IFC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the IFC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFC, as well as average returns and alphas for long/short trading IFC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IFC strategy achieves an average return of 39 bps/month

with a t-statistic of 4.43. Among these large cap stocks, the alphas for the IFC strategy relative to the five most common factor models range from 31 to 45 bps/month with t-statistics between 3.49 and 5.02.

5 How does IFC perform relative to the zoo?

Figure 2 puts the performance of IFC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IFC strategy falls in the distribution. The IFC strategy’s gross (net) Sharpe ratio of 0.59 (0.47) is greater than 95% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IFC strategy (red line).² Ignoring trading costs, a \$1 invested in the IFC strategy would have yielded \$4.37 which ranks the IFC strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFC strategy would have yielded \$2.73 which ranks the IFC strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IFC relative to those. Panel A shows that the IFC strategy gross alphas fall between the 65 and 72 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IFC strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFC ranks between the 79 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IFC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IFC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IFC or at least to weaken the power IFC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC}IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IFC signal in these Fama-MacBeth regressions exceed 1.76, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on IFC is 0.71.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IFC strategy earns alphas that range from 19-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.86, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFC trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.90.

7 Does IFC add relative to the whole zoo?

Finally, we can ask how much adding IFC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the IFC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes IFC grows to \$1241.53.

8 Conclusion

This study provides robust evidence that Investment Financing Conservatism (IFC) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that IFC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IFC delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.47

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IFC is available.

and statistically significant abnormal returns of 17 basis points per month relative to the Fama-French five-factor model plus momentum (t -statistic = 2.57). Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a significant alpha of 20 basis points per month (t -statistic = 2.90) even after accounting for six closely related financing and investment strategies. This robustness suggests that IFC captures unique information about firm fundamentals not fully explained by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, IFC adds to our understanding of how firms’ financing decisions relate to expected returns, supporting theories that link conservative financing policies to systematic risk exposures or behavioral biases. For practitioners, the strategy’s net Sharpe ratio of 0.47 indicates that IFC remains economically viable even after accounting for realistic transaction costs, suggesting potential value for quantitative investment strategies.

Several limitations of this study warrant consideration. First, while we control for numerous related factors, the possibility remains that IFC proxies for an omitted risk factor or market friction not captured in our analysis. Second, the implementation of IFC strategies in practice may face capacity constraints and market impact costs beyond those captured in our net return calculations. Third, our sample period and geographic focus may limit the generalizability of results to other markets or time periods.

Future research should explore several promising directions. Investigation into the economic mechanisms driving the IFC premium would enhance our theoretical understanding of this anomaly. International evidence could establish whether IFC’s predictive power extends beyond U.S. markets. Additionally, examining the interaction between IFC and firm characteristics such as size, profitability, or financial constraints could reveal important heterogeneity in the signal’s effectiveness. Finally,

investigating the time-variation in IFC's performance and its relationship to macroeconomic conditions could provide insights into when the strategy is most effective and whether it represents compensation for systematic risk or market inefficiency.

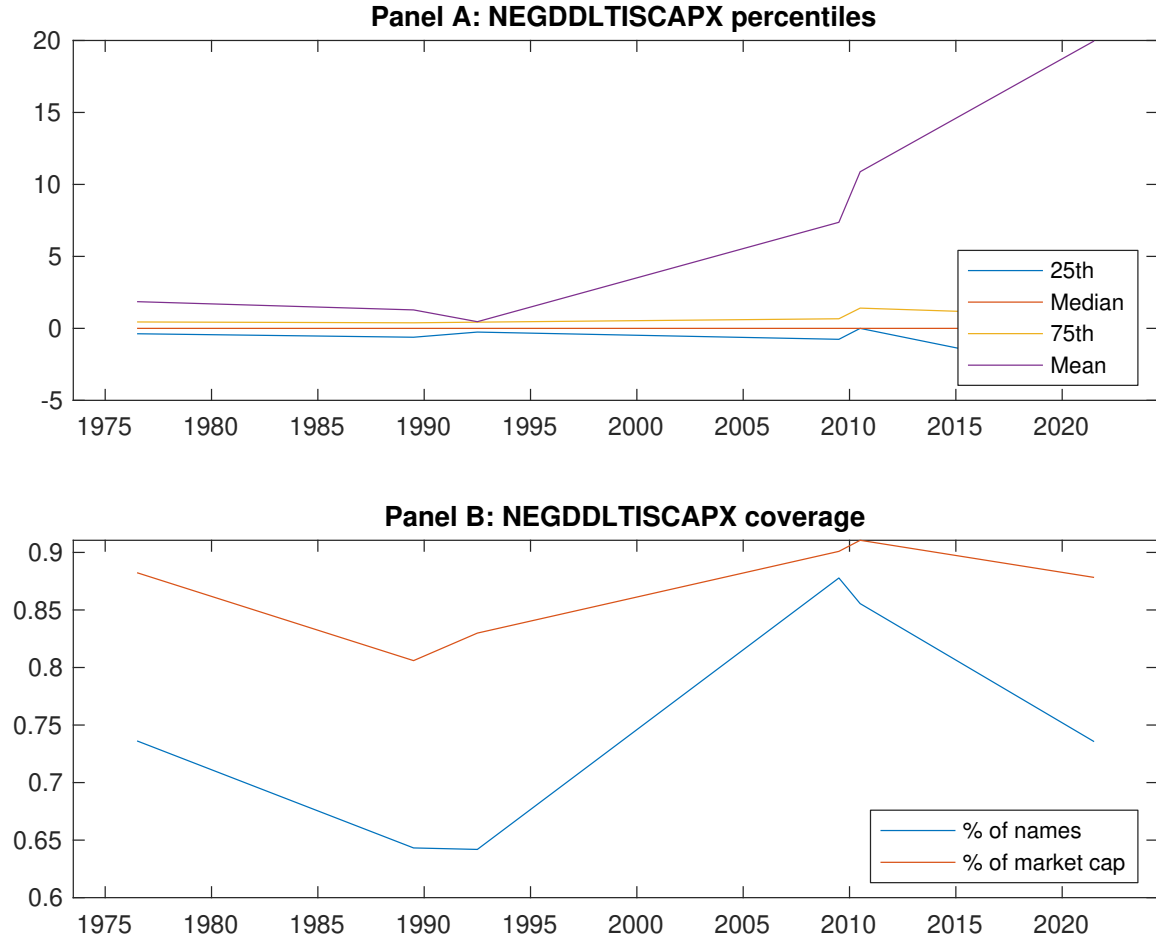


Figure 1: Times series of IFC percentiles and coverage. This figure plots descriptive statistics for IFC. Panel A shows cross-sectional percentiles of IFC over the sample. Panel B plots the monthly coverage of IFC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on IFC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.56]	0.67 [3.76]	0.65 [3.26]	0.72 [4.05]	0.84 [4.19]	0.29 [4.26]
α_{CAPM}	-0.18 [-3.44]	0.07 [1.35]	-0.02 [-0.33]	0.12 [2.35]	0.16 [3.00]	0.34 [5.05]
α_{FF3}	-0.20 [-3.91]	0.04 [0.97]	0.05 [0.93]	0.12 [2.38]	0.13 [2.57]	0.34 [4.95]
α_{FF4}	-0.17 [-3.21]	0.04 [0.91]	0.10 [1.86]	0.07 [1.44]	0.13 [2.40]	0.30 [4.30]
α_{FF5}	-0.16 [-3.10]	-0.03 [-0.77]	0.10 [1.75]	0.03 [0.65]	0.08 [1.48]	0.24 [3.54]
α_{FF6}	-0.14 [-2.65]	-0.03 [-0.68]	0.13 [2.41]	0.00 [0.06]	0.08 [1.47]	0.22 [3.18]
Panel B: Fama and French (2018) 6-factor model loadings for IFC-sorted portfolios						
β_{MKT}	1.08 [89.35]	0.98 [95.69]	0.97 [75.06]	0.96 [83.22]	1.05 [84.68]	-0.03 [-1.99]
β_{SMB}	0.11 [6.08]	-0.12 [-7.63]	-0.02 [-0.83]	-0.02 [-1.12]	0.11 [6.04]	0.00 [0.09]
β_{HML}	0.08 [3.42]	0.06 [2.92]	-0.17 [-6.82]	-0.08 [-3.57]	-0.02 [-0.83]	-0.10 [-3.28]
β_{RMW}	0.01 [0.36]	0.14 [7.09]	-0.04 [-1.40]	0.08 [3.53]	0.08 [3.09]	0.07 [2.15]
β_{CMA}	-0.16 [-4.48]	0.10 [3.20]	-0.10 [-2.76]	0.22 [6.55]	0.13 [3.62]	0.29 [6.28]
β_{UMD}	-0.04 [-3.03]	-0.01 [-0.59]	-0.06 [-4.54]	0.05 [4.10]	-0.00 [-0.02]	0.04 [2.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	674	504	1021	562	636	
me (\$10 ⁶)	1502	2808	2233	2766	1568	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IFC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [4.26]	0.34 [5.05]	0.34 [4.95]	0.30 [4.30]	0.24 [3.54]	0.22 [3.18]
Quintile	NYSE	EW	0.19 [4.18]	0.21 [4.65]	0.20 [4.38]	0.19 [4.13]	0.18 [4.01]	0.18 [3.91]
Quintile	Name	VW	0.26 [3.58]	0.31 [4.26]	0.33 [4.46]	0.29 [3.88]	0.26 [3.63]	0.24 [3.31]
Quintile	Cap	VW	0.31 [4.74]	0.35 [5.46]	0.36 [5.68]	0.31 [4.82]	0.28 [4.41]	0.25 [3.87]
Decile	NYSE	VW	0.32 [3.64]	0.37 [4.16]	0.37 [4.21]	0.36 [4.04]	0.24 [2.78]	0.25 [2.85]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.36]	0.28 [4.07]	0.27 [3.93]	0.25 [3.61]	0.19 [2.81]	0.17 [2.57]
Quintile	NYSE	EW	-0.08 [-1.37]					
Quintile	Name	VW	0.20 [2.74]	0.24 [3.28]	0.25 [3.44]	0.23 [3.14]	0.21 [2.80]	0.19 [2.57]
Quintile	Cap	VW	0.25 [3.92]	0.29 [4.48]	0.30 [4.65]	0.27 [4.22]	0.23 [3.59]	0.20 [3.28]
Decile	NYSE	VW	0.25 [2.79]	0.29 [3.21]	0.29 [3.23]	0.29 [3.17]	0.18 [2.06]	0.18 [2.02]

Table 3: Conditional sort on size and IFC

This table presents results for conditional double sorts on size and IFC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFC and short stocks with low IFC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IFC Quintiles					IFC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.72 [2.65]	0.90 [3.26]	0.91 [3.25]	0.95 [3.32]	0.77 [2.84]	0.05 [0.70]	0.07 [0.90]	0.05 [0.70]	0.05 [0.65]	0.08 [1.00]	0.08 [0.97]
	(2)	0.81 [3.06]	0.81 [3.20]	0.87 [3.41]	0.91 [3.64]	0.81 [3.24]	0.01 [0.08]	0.04 [0.52]	0.00 [0.04]	0.01 [0.16]	-0.04 [-0.50]	-0.03 [-0.36]
	(3)	0.74 [3.01]	0.85 [3.76]	0.80 [3.30]	0.79 [3.48]	0.93 [3.98]	0.19 [2.57]	0.23 [3.17]	0.22 [3.03]	0.18 [2.37]	0.21 [2.85]	0.18 [2.38]
	(4)	0.74 [3.34]	0.80 [3.76]	0.82 [3.71]	0.78 [3.63]	0.88 [4.07]	0.14 [1.88]	0.16 [2.19]	0.14 [1.93]	0.12 [1.68]	0.09 [1.26]	0.09 [1.15]
	(5)	0.45 [2.20]	0.63 [3.55]	0.59 [3.05]	0.62 [3.41]	0.84 [4.34]	0.39 [4.43]	0.42 [4.78]	0.45 [5.02]	0.37 [4.21]	0.36 [4.01]	0.31 [3.49]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IFC Quintiles					IFC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	380	381	381	381	377	35	34	31	33	34	
	(2)	105	105	105	105	105	60	60	59	60	59	
	(3)	75	75	75	75	75	106	106	103	104	106	
	(4)	63	63	63	63	62	226	234	225	230	224	
(5)	57	57	57	57	57	1328	2082	1748	2145	1446		

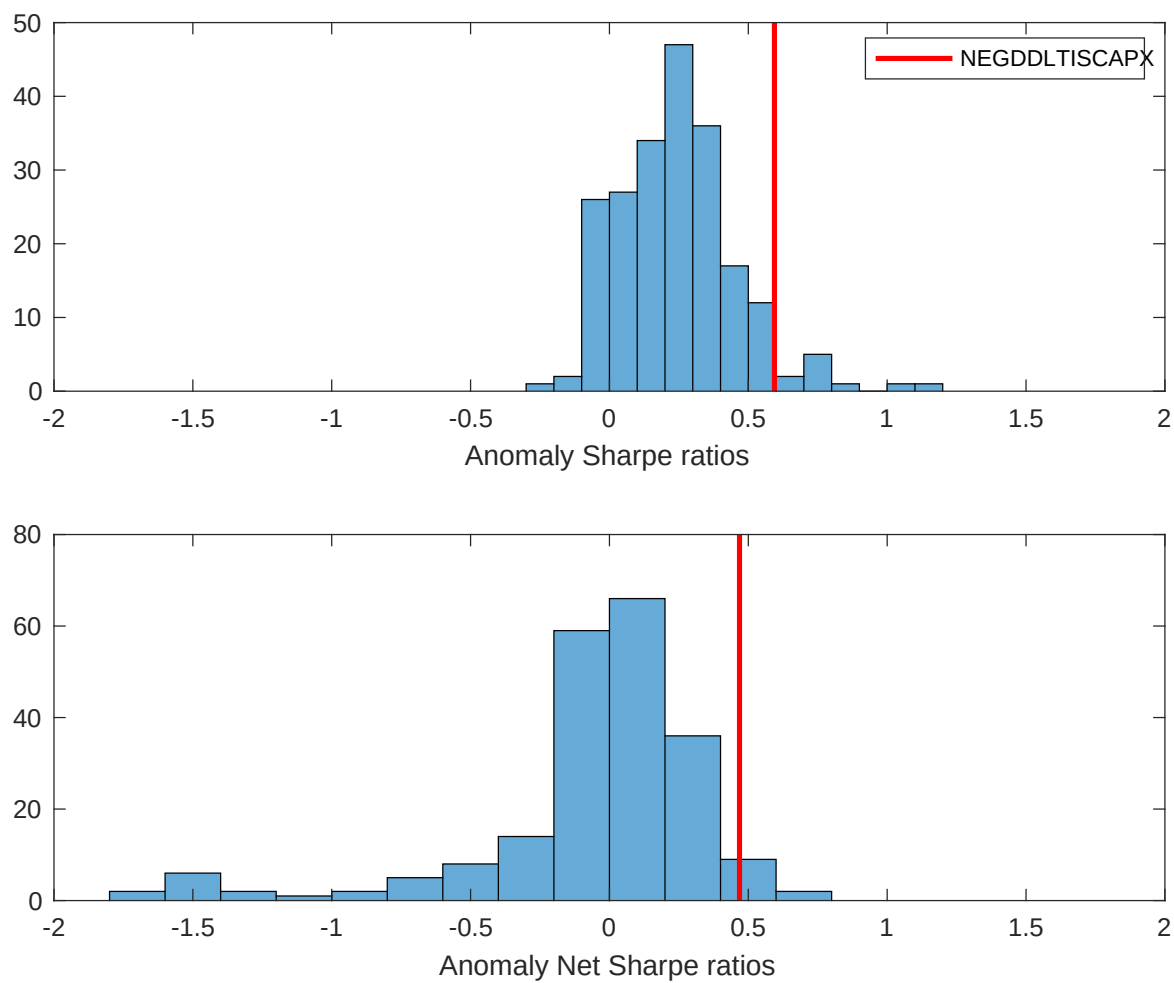


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

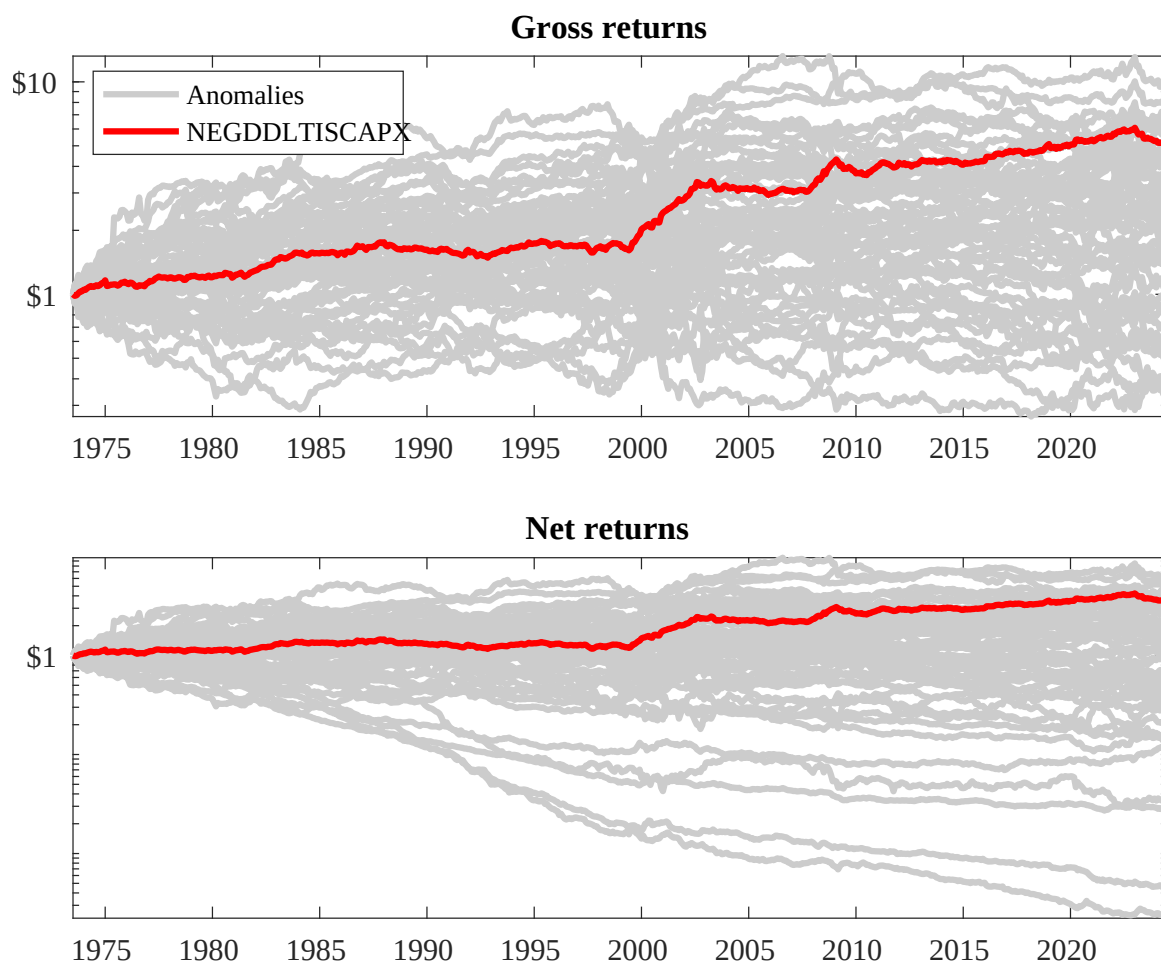
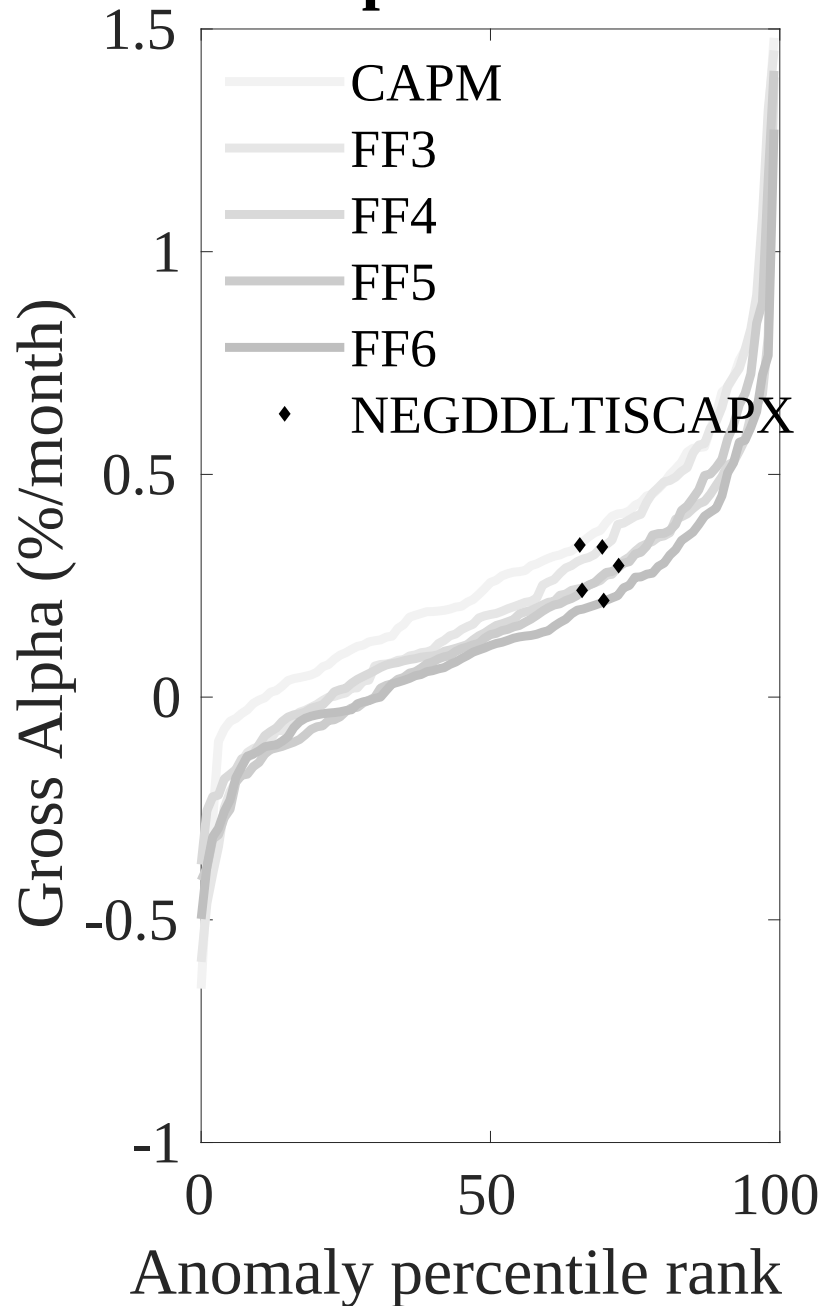


Figure 3: Dollar invested.

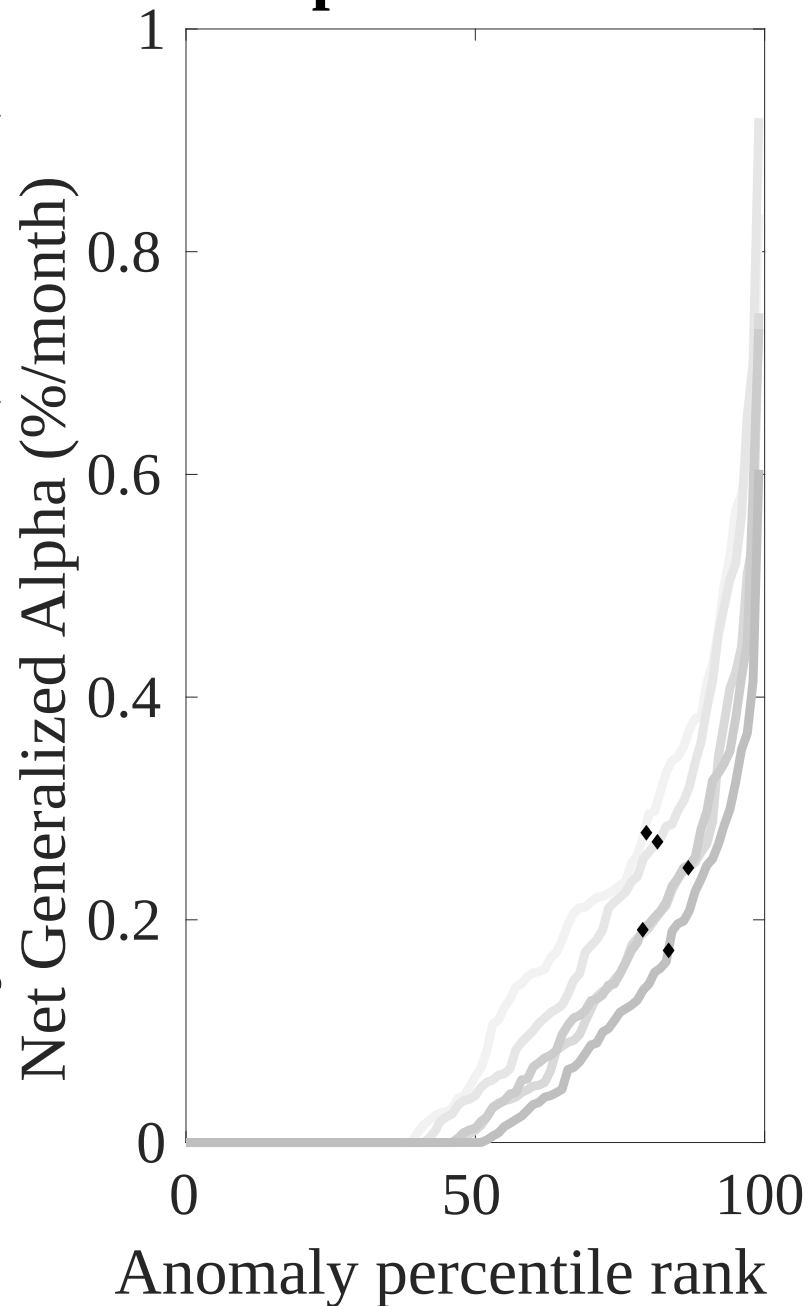
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



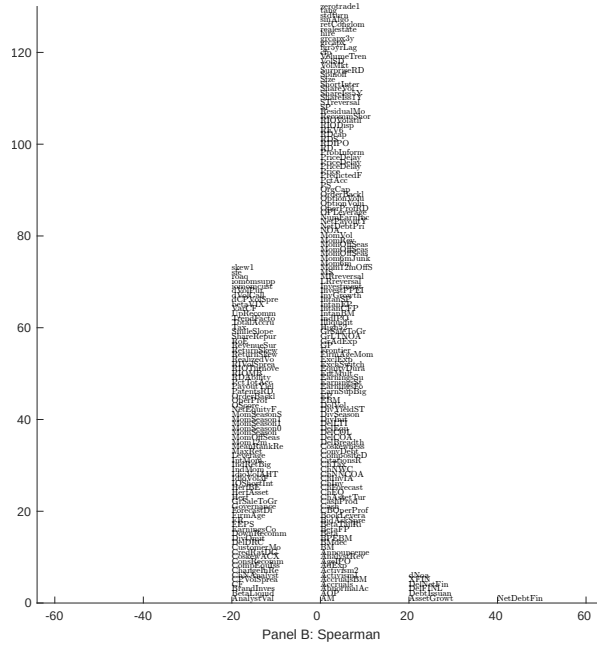
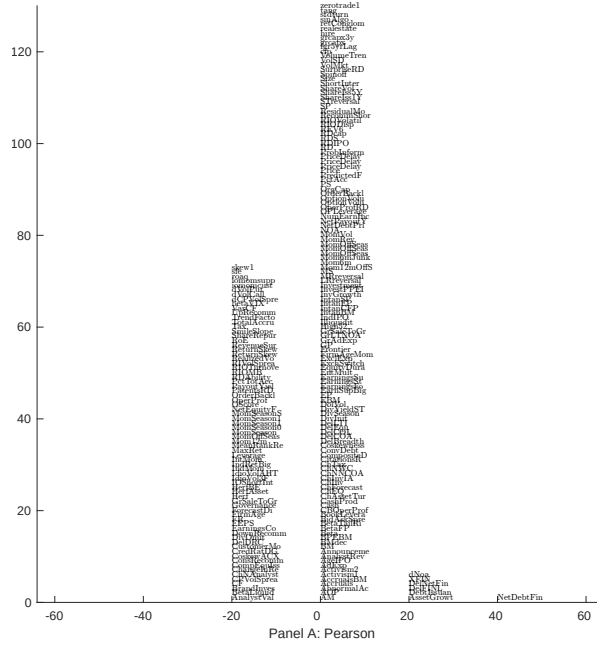


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with IFC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

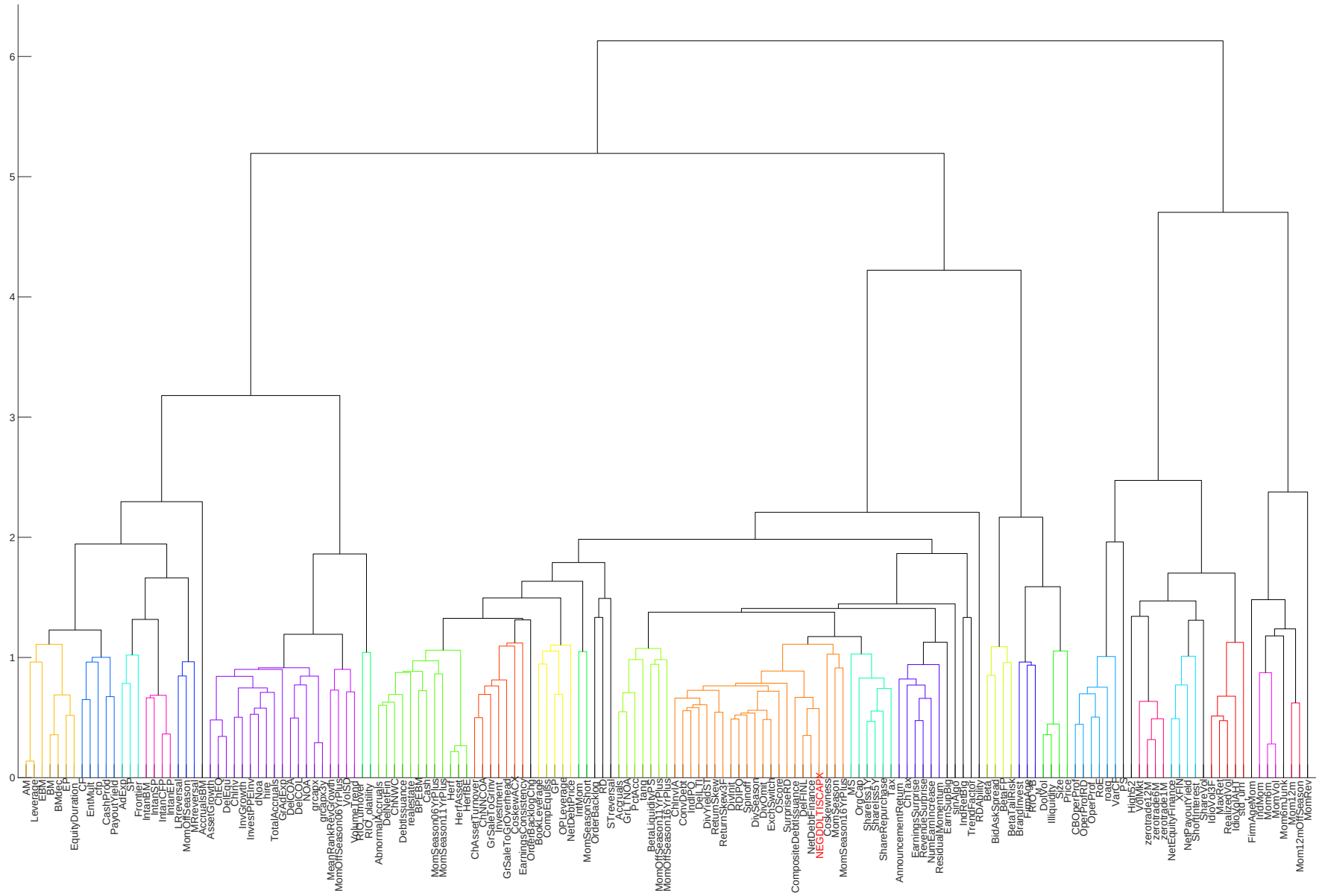


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

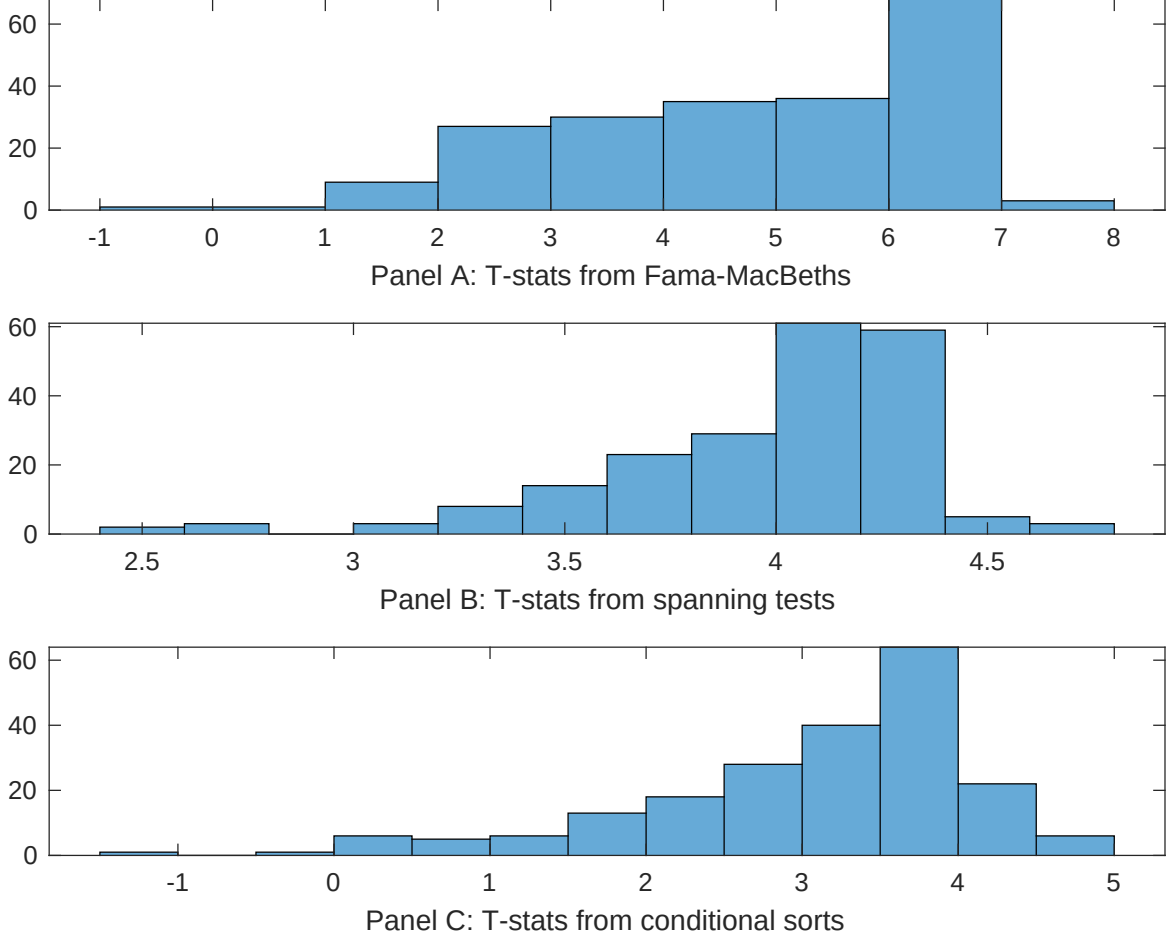


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC}IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IFC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFC} IFC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.38]	0.14 [5.69]	0.13 [5.34]	0.14 [5.81]	0.14 [5.68]	0.13 [5.35]	0.15 [5.96]
IFC	0.51 [2.10]	0.80 [3.11]	0.61 [2.32]	0.54 [2.05]	0.45 [1.76]	0.13 [5.28]	0.19 [0.71]
Anomaly 1	0.17 [9.10]						-0.54 [-1.32]
Anomaly 2		0.20 [6.43]					0.80 [1.53]
Anomaly 3			0.21 [9.11]				0.60 [1.06]
Anomaly 4				0.11 [8.61]			0.49 [2.64]
Anomaly 5					0.14 [9.67]		0.73 [4.55]
Anomaly 6						0.95 [6.20]	0.12 [0.98]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IFC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.86]	0.20 [2.89]	0.20 [2.93]	0.21 [3.05]	0.21 [3.01]	0.22 [3.18]	0.20 [2.90]
Anomaly 1	14.96 [3.91]						11.09 [2.10]
Anomaly 2		10.79 [3.18]					7.41 [2.02]
Anomaly 3			15.30 [4.15]				7.07 [1.40]
Anomaly 4				8.20 [2.08]			6.10 [1.41]
Anomaly 5					1.70 [0.43]		-7.28 [-1.66]
Anomaly 6						3.45 [0.94]	2.09 [0.54]
mkt	-2.57 [-1.65]	-1.43 [-0.88]	-2.88 [-1.86]	-2.91 [-1.86]	-2.91 [-1.85]	-2.82 [-1.79]	-1.59 [-0.97]
smb	-0.73 [-0.31]	4.27 [1.64]	-0.55 [-0.23]	0.21 [0.09]	0.84 [0.35]	1.04 [0.43]	0.99 [0.36]
hml	-8.39 [-2.79]	-8.18 [-2.70]	-8.76 [-2.93]	-9.77 [-3.23]	-9.51 [-3.12]	-10.04 [-3.22]	-7.62 [-2.45]
rmw	5.31 [1.73]	0.21 [0.06]	4.97 [1.62]	6.68 [2.18]	6.68 [2.17]	6.81 [2.21]	0.45 [0.12]
cma	23.78 [5.04]	21.75 [4.29]	24.82 [5.35]	18.86 [2.85]	27.52 [5.08]	25.70 [4.55]	14.55 [2.08]
umd	2.47 [1.56]	3.69 [2.38]	2.78 [1.78]	3.93 [2.52]	3.71 [2.37]	3.54 [2.24]	2.52 [1.59]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	13	14	12	11	11	15

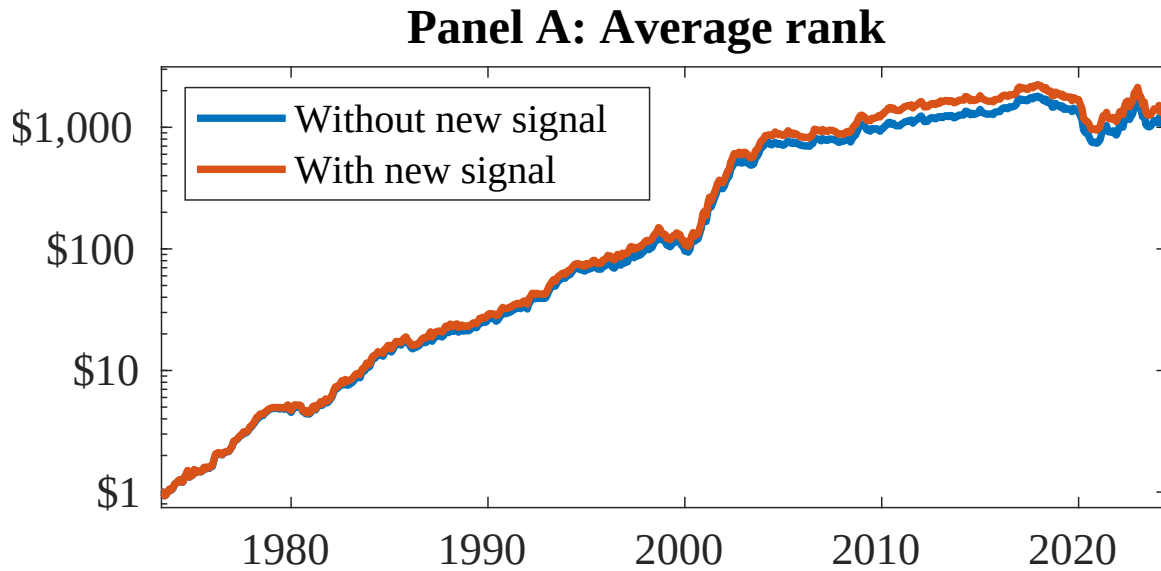


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as IFC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cochrane, J. H. (2005). *Asset Pricing*. Princeton University Press, revised edition.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Hansen, L. P., Heaton, J. C., and Li, N. (2008). Consumption strikes back? measuring long-run risk. *Journal of Political Economy*, 116(2):260–302.
- Hansen, L. P. and Singleton, K. J. (1982). Generalized instrumental variables estimation of nonlinear rational expectations models. *Econometrica*, 50(5):1269–1286.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.